

Continuous Convergence and Curried Functional Spaces

1. The language of filters

Definition 1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 1.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ generated by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.

If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x .

If $f : X \rightarrow Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X , then $\{f(A) \mid A \in F\}$ is a filter base that generates the *image filter* denoted $f(F)$.

Definition 1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \rightarrow x$, if it contains all open neighborhoods of x .

Remark 1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter

$$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$

converges to x . This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.

- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \emptyset$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X .
- A function $f : X \rightarrow Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \rightarrow x \in X$ implies $f(F) \rightarrow f(x) \in Y$.

2. Curried functions

Definition 2.1. Let $n \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$. The set $H_n(\mathbb{R})$ of *curried n -variable functions* is a topological vector space defined as follows:

- If $n = 0$, we set $H_0(\mathbb{R}) = \mathbb{R}$;
- If $H_n(\mathbb{R})$ is defined, we define $H_{n+1}(\mathbb{R})$ to be the set of all functions from $\mathbb{R}$ to $H_n(\mathbb{R})$. This set is given the pointwise topological and linear structure. In other words, a filter $F \in \mathcal{F}(H_{n+1}(\mathbb{R}))$ converges to $f \in H_{n+1}(\mathbb{R})$ if and only if $F(x) = \{A(x) \mid A \in F\}$ converges to $f(x)$ for all $x \in \mathbb{R}$. It is not hard to see that this topology is compatible with the pointwise vector structure.

Definition 2.2. Let $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function. Define

$$\gamma(\tilde{f})(x_1)(x_2)\dots(x_n) = \tilde{f}(x_1, x_2, \dots, x_n).$$

Then $\gamma : \text{Hom}(\mathbb{R}^n, \mathbb{R}) \rightarrow H_n(\mathbb{R})$ is clearly a bijection. The function $f = \gamma(\tilde{f})$ will be called the *curried version* of $\tilde{f}$.

3. Continuous functions

Definition 3.1. One may try to decouple the notion of convergent filters from topology. Let X be a set with a relation $\rightarrow \subset \mathcal{F}(X) \times X$. Then $(X, \rightarrow)$ is called a *convergence space* if

- For all $x \in X$, we have $[x] \rightarrow x$;
- If $F \rightarrow x$ and $G \rightarrow x$, then $F \cap G \rightarrow x$;
- If $F \rightarrow x$ and $F \subset G$, then $G \rightarrow x$.

Definition 3.2. A map $f : X \rightarrow Y$ between convergence spaces is said to be *continuous* if $f(F) \rightarrow f(x) \in Y$ whenever $F \rightarrow x \in X$.

Definition 3.3. Let X, Y be two convergence spaces. Then the product $X \times Y$ is endowed with the following convergence structure: a filter $F \in \mathcal{F}(X \times Y)$ converges to $(x, y) \in X \times Y$ if and only if $p_X(F) \rightarrow x \in X$ and $p_Y(F) \rightarrow y \in Y$.

Definition 3.4. (see [2], pages 25-26): Let X, Y be convergence spaces. By $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y . Then $\mathcal{C}(X, Y)$ can be endowed with a convergence structure as follows. For a filter $F \in \mathcal{F}(\mathcal{C}(X, Y))$, we say that $F \rightarrow f \in \mathcal{C}(X, Y)$ if and only if $F(P) \rightarrow f(p)$ whenever $P \rightarrow p \in X$. Here, the filter $F(P)$ is based on

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

The resulting convergence structure on $\mathcal{C}(X, Y)$ is called the structure of *continuous convergence*.

Remark 3.1. Continuous convergence on $\mathcal{C}(X, Y)$ is the coarsest convergence structure that makes the *evaluation mapping*

$$\omega : \mathcal{C}(X, Y) \times X \rightarrow Y$$

continuous. Here, the product $\mathcal{C}(X, Y) \times X$ is endowed with the product convergence structure as per [Definition 3.3](#).

Proposition 3.1. (see [2], page 27): *Let X, Y, Z be convergence spaces. Then the map*

$$\gamma : \mathcal{C}(X \times Y, Z) \rightarrow \mathcal{C}(X, \mathcal{C}(Y, Z))$$

defined as $\gamma(f)(x)(y) = f(x, y)$, is well-defined and is a homeomorphism.

Proposition 3.2. (see [2], page 34): *Let X, Y be topological spaces such that X is locally compact and Y is regular. Then the convergence space $\mathcal{C}(X, Y)$ is topologized by the compact-open topology (see [3]). That is, the convergence relation arising from this topology coincides precisely with the convergence structure given in [Definition 3.4](#).*

Proposition 3.3. (see [2], page 28): *Let X be a convergence space and let Y be a Hausdorff convergence space (i.e. every filter in Y converges to at most one point). Then $\mathcal{C}(X, Y)$ is also a Hausdorff convergence space.*

Proposition 3.4. (see [2], page 29): *Let X be a convergence space and Y be a convergence vector space, i.e. a set endowed with both structures such that the operations $+: Y \times Y \rightarrow Y$ and $\cdot: \mathbb{R} \times Y \rightarrow Y$ are continuous. Then $\mathcal{C}(X, Y)$ is also a convergence vector space.*

Corollary 3.1. *If X is a locally compact topological space and Y is a regular topological vector space, then $\mathcal{C}(X, Y)$ is a topological vector space.*

Proposition 3.5. *Let X, Y be topological spaces such that X is first countable. Then a sequence $\{f_k\}_{k \in \mathbb{N}} \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ (in the sense that its derived filter converges) if and only if $f_k(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$ whenever $x_k \xrightarrow[k \rightarrow \infty]{} x$ in X .*

Proof: First, let F be the derived filter of $\{f_k\}_{k \in \mathbb{N}}$ and suppose that F converges to a function $f \in \mathcal{C}(X, Y)$. Consider a sequence $\{x_k\}_{k \in \mathbb{N}}$ converging to $x \in X$, with its derived filter P . Now let V be a neighborhood of $f(x)$ in Y . By the definition of continuous convergence, we have $F(P) \rightarrow f(x)$, and so $V \in F(P)$. This means that V contains an element of the base of $F(P)$, namely a set

$$B_{n_0, m_0} = \{f_n(x_m) \mid n \geq n_0, m \geq m_0\}.$$

Hence, for $k \geq \max(n_0, m_0)$, we have $f_k(x_k) \in V$, as desired.

Conversely, suppose that $f_k(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$ whenever $x_k \xrightarrow[k \rightarrow \infty]{} x \in X$. First let us show that for any subsequence $\{f_{n_k}\}_{k \in \mathbb{N}}$, we have $f_{n_k}(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$.

To this end, let $n \in \mathbb{N}$. We let $z_n = x_{\varphi(n)}$, where $\varphi(n)$ is the minimal number of those $k \in \mathbb{N}$ that satisfy $n \leq n_k$. We clearly see that $n_1 \geq n_2$ implies $\varphi(n_1) \geq \varphi(n_2)$ and that $\varphi(n_k) = k$ for all $k \in \mathbb{N}$. Therefore, the sequence $\{z_n\}_{n \in \mathbb{N}}$ converges to x . Hence we have $f_k(z_k) \xrightarrow[k \rightarrow \infty]{} f(x)$, and so

$$f_{n_k}(x_k) = f_{n_k}(z_{n_k}) \xrightarrow[k \rightarrow \infty]{} f(x). \quad (1)$$

Now, to show that $f_k \rightarrow f$ in $\mathcal{C}(X, Y)$, we need to show that the derived filter F generated by the sets $C_n = \{f_k \mid k \geq n\}$ converges to f . To this end, let P be a filter in X converging to a point $x \in X$. Note that since X is first countable, the filter P has a countable base $\{A_m\}_{m \in \mathbb{N}}$ of neighborhoods of x . To show that $F(P)$ converges to $f(x)$, assume the contrary. Then some neighborhood V of $f(x)$ does not lie in $F(P)$, which is to say that for all $n, m \in \mathbb{N}$, we have $C_n(A_m) \not\subset V$. That means that for all $n, m \in \mathbb{N}$, there are $k_m \geq m$ and $x_m \in A_m$ such that $f_{k_m}(x_m) \notin V$.

In particular, we have $f_{k_n}(x_n) \notin V$ for all $n \in \mathbb{N}$. At the same time, since the sets A_n generate P , we see that $x_n \xrightarrow[n \rightarrow \infty]{} x$, so we must have $f_{k_n}(x_n) \xrightarrow[n \rightarrow \infty]{} f(x)$ by (1), a contradiction. ■

Definition 3.5. Let $n \in \mathbb{N}_0$. The set $C_n(\mathbb{R})$ of *curried continuous functions* is a Hausdorff topological vector space defined recursively as follows:

- If $n = 0$, we set $C_0(\mathbb{R}) = \mathbb{R}$.
- If $C_n(\mathbb{R})$ is defined, we set $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$. Proposition 3.2, Proposition 3.3, and Corollary 3.1 ensure that $C_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space, provided that $C_n(\mathbb{R})$ is.

Proposition 3.6. Let $n \in \mathbb{N}_0$. Then the map

$$\gamma : \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \rightarrow C_n(\mathbb{R})$$

defined by $\gamma(f)(x_1)(x_2)\dots(x_n) = f(x_1, \dots, x_n)$, is well-defined and is a homeomorphism.

Proof: If $n = 0$, the statement is trivial. Assume it is proven for $C_n(\mathbb{R})$. By

Proposition 3.1, we have

$$\mathcal{C}(\mathbb{R}^{n+1}, \mathbb{R}) = \mathcal{C}(\mathbb{R} \times \mathbb{R}^n, \mathbb{R}) \cong \mathcal{C}(\mathbb{R}, \mathcal{C}(\mathbb{R}^n, \mathbb{R})) \cong \mathcal{C}(\mathbb{R}, C_n(\mathbb{R})) = C_{n+1}(\mathbb{R}),$$

q.e.d. ■

4. Differentiable functions

Definition 4.1. Let $n \in \mathbb{N}$. A curried function $f \in H_n(\mathbb{R})$ is called *differentiable* at

$(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ if

- $f(x_1) \in H_{n-1}(\mathbb{R})$ is differentiable at $(x_2, \dots, x_n) \in \mathbb{R}^{n-1}$, if $n \geq 2$;
- There exists an element $f'(x_1) \in H_{n-1}(\mathbb{R})$ such that

$$f(x_1 + h) = f(x_1) + h \cdot f'(x_1) + o(h),$$

where $o(h) \in H_{n-1}(\mathbb{R})$ is a function such that $\frac{o(h)}{h} \rightarrow 0$ as $h \rightarrow 0$. Equivalently, we may write

$$f'(x_1) = \lim_{h \rightarrow 0} \frac{f(x_1 + h) - f(x_1)}{h}.$$

Proposition 4.1. A curried function $f \in H_n(\mathbb{R})$ is differentiable at $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ if and only if $\tilde{f} = \gamma^{-1}(f)$ has all partial derivatives at $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

Proof: We prove by induction over n . If $n = 1$, the statement is trivial, so assume $n \geq 2$ and the statement proven for $1, 2, \dots, n-1$.

Suppose that $f \in H_n$ is differentiable. Then for each $i \geq 2$, we have

$$\frac{\partial \tilde{f}}{\partial x_i}(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{\tilde{f}(x_1, \dots, x_{i-1}, x_i + h, x_{i+1}, \dots, x_n) - \tilde{f}(x_1, \dots, x_n)}{h} \quad (2)$$

$$= \frac{\partial \gamma^{-1}(f(x_1)(x_2)\dots(x_{i-1}))}{\partial x_i}(x_i, x_{i+1}, \dots, x_n), \quad (3)$$

which exists by the induction hypothesis, since $f(x_1)\dots(x_i) \in H_{n-i}(\mathbb{R})$ is differentiable.

It remains to show that $\frac{\partial \tilde{f}}{\partial x_1}$ exists. But by the definition of $f'(x_1)$, we have

$$f'(x_1) = \lim_{h \rightarrow 0} \frac{f(x_1 + h) - f(x_1)}{h}. \quad (4)$$

Since the limit is taken in $H_n(\mathbb{R})$, by applying (4) to $x_2, x_3, \dots, x_n$, we get

$$\frac{\partial \tilde{f}}{\partial x_1}(x_1, x_2, \dots, x_n) = f'(x_1)(x_2)\dots(x_n).$$

Now suppose that $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{R}$ has all partial derivatives at $(x_1, \dots, x_n)$. By (2), we see that $\tilde{f}(x_1) : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ has all partial derivatives at $(x_2, x_3, \dots, x_n)$, and so by the induction hypothesis $f(x_1)$ is differentiable. Finally, we set $f'(x_1) := \gamma\left(\frac{\partial \tilde{f}}{\partial x_1}\right)(x_1)$, and easily see that (4) holds for $f'(x_1)$ so defined. Therefore f is differentiable. ■

Definition 4.2. A curried function $f \in H_n(\mathbb{R})$ is called *continuously differentiable* if f is differentiable everywhere, and the maps $f', x \mapsto f(x)'$ are continuous.

Proposition 4.2. If $f \in H_n(\mathbb{R})$ is continuously differentiable, then f is continuous.

Proposition 4.3. A curried function $f \in H_n(\mathbb{R})$ is continuously differentiable if and only if $\tilde{f} = \gamma^{-1}(f)$ has continuous partial derivatives everywhere on $\mathbb{R}^n$.

Proof: The base case $n = 1$ is trivial, so assume the statement proven for $1, 2, \dots, n - 1$. We again follow two steps.

First suppose that $f \in H_n(\mathbb{R})$ is continuously differentiable. By [Proposition 4.1](#) we see that $\tilde{f}$ has partial derivatives everywhere on $\mathbb{R}^n$, so it remains to show these derivatives are continuous. With respect to x_1 , we have

$$\frac{\partial \tilde{f}}{\partial x_1} = \gamma^{-1}(f')$$

which is continuous since f' is continuous. To show the continuity of $\frac{\partial \tilde{f}}{\partial x_2}$, consider a sequence $x^{(k)} = (x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)}) \in \mathbb{R}^n$, converging to a point $x = (x_1, \dots, x_n)$. By the continuity of the map $t \mapsto f(t)'$, we have $f(x_1^{(k)})' \rightarrow f(x_1)'$ in $C_{n-1}(\mathbb{R})$, which implies

$$\frac{\partial \tilde{f}}{\partial x_2}(x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)}) = f(x_1^{(k)})'(x_2^{(k)}) \dots (x_n^{(k)}) \xrightarrow{k \rightarrow \infty} f(x_1)'(x_2) \dots (x_n) = \frac{\partial \tilde{f}}{\partial x_2}(x_1, \dots, x_n),$$

so $\frac{\partial \tilde{f}}{\partial x_2}$ is continuous. ■

Bibliography

- [1] N. Bourbaki, *General Topology, Part I*. 1966.
- [2] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002.
- [3] R. H. Fox, "On Topologies For Function Spaces," 1945.